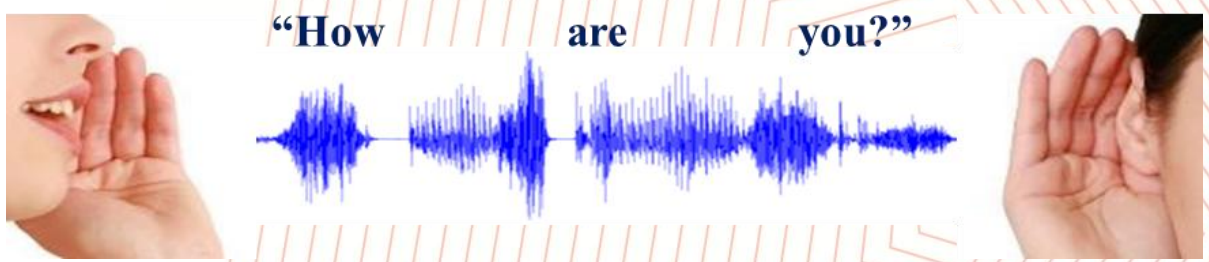
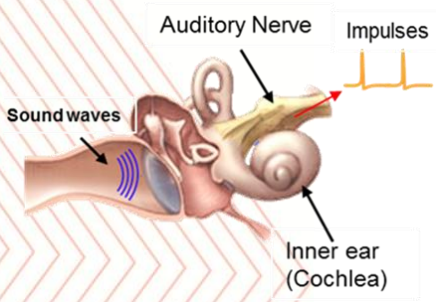
ELEC3104: Mini-Project – Cochlear Signal Processing





Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

Higher-Level: Project Implementation

- ✓ Use the cascaded cochlear model developed in the pass-level project to complete the following tasks:
  - **Task 1 – Short-Time Spectrum Analysis:** Incorporate an inner hair cell model into the cascaded cochlear model to develop a short-time spectrum analyser. Test its performance using sinusoidal signals and compare the results with the FFT magnitude spectrum.
  - **Task 2 – Pitch Period Estimation:** Use the spectrum analyser to estimate the fundamental frequency and pitch period of voiced speech. Compare the results with manual estimates from the speech waveform.
  - **Task 3 – Missing Fundamental:** Investigate pitch estimation when the fundamental frequency component is removed by band-pass filtering. Compare the original and filtered speech and explain the results.

ELEC3104: Mini-Project – Cochlear Signal Processing

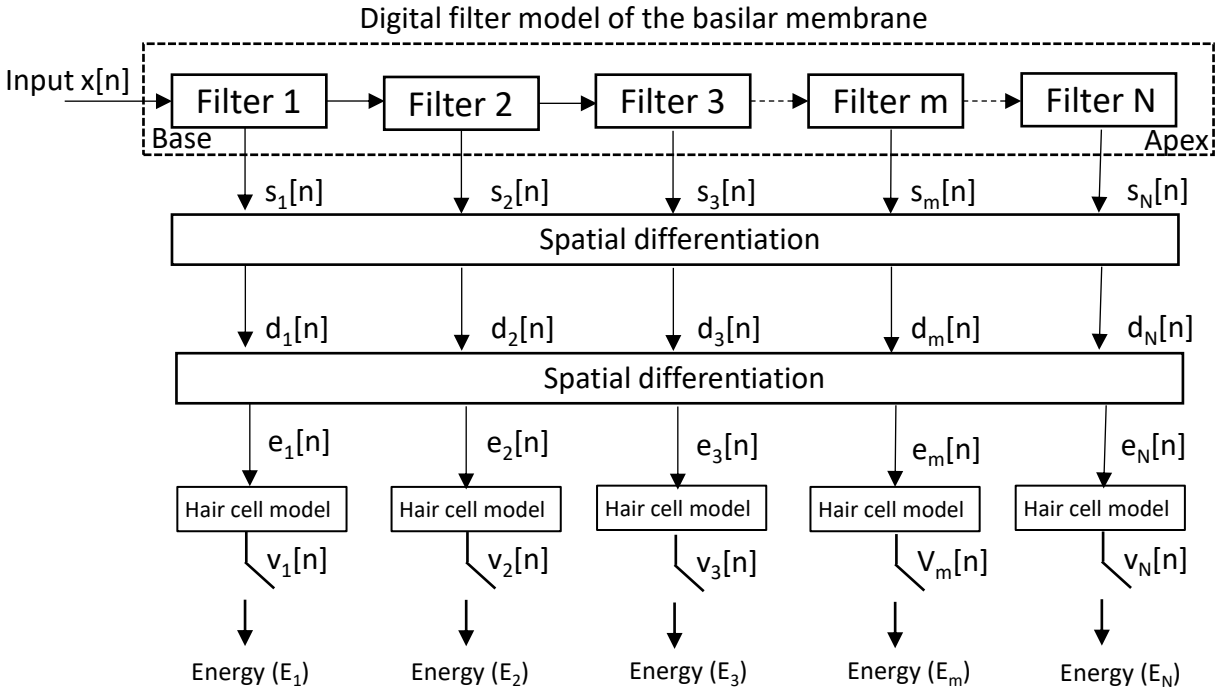
Higher –Level (Task1): Short-Time Spectrum Analysis Using the Cascaded Cochlear Model

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

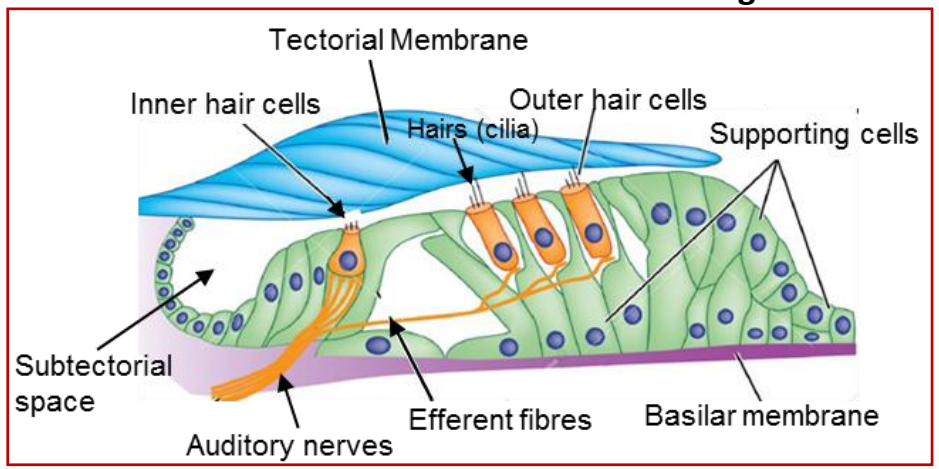
Short-time Spectrum Analyser

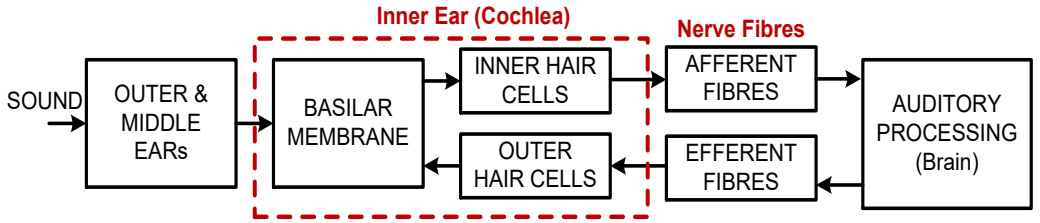
- Use the cochlear model developed in the pass-level task to implement a short-time spectrum analyser. The cascaded filters divide the frequency range of interest (35–7500 Hz) into (N = 512) frequency bands.
- Apply two stages of spatial differentiation to sharpen the magnitude responses of the filters.
- Pass each spatially differentiated output through a hair cell model consisting of a rectifier followed by a first-order low-pass filter. The resulting outputs provide a short-time representation of the signal level (measure of Energy) across the frequency bands.



Ouput Energy E_m is read (switch closed) once every 16 ms or so.

Organ of Corti





Inner hair cell model

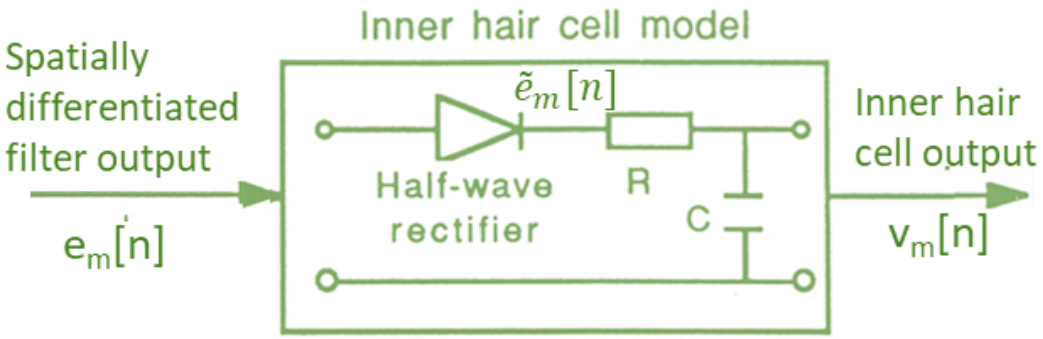
Inner hair cell model

The inner hair cell is represented by a half-wave rectifier followed by an RC low-pass filter. The input voltage represents the spatially differentiated basilar membrane displacement from the filter bank.

- ✓ Bending the inner hair cell cilia (Half-wave rectification) is simulated by charging of the capacitor and returning to the initial position of the cilia is equivalent to discharging the capacitor.
- ✓ This model is given by the following input-output relationship:

$$v_m[n] = (1 - c_0)\tilde{e}_m[n] + c_0 v_m[n-1] \quad \text{where } c_0 = e^{-2\pi\frac{f_c}{f_s}}$$

Where, $v_m(n)$ is the output electrical energy.

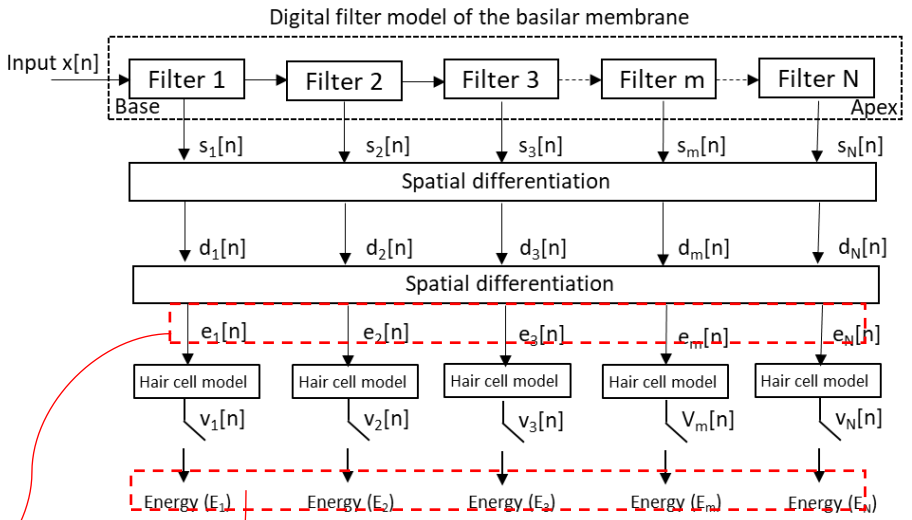


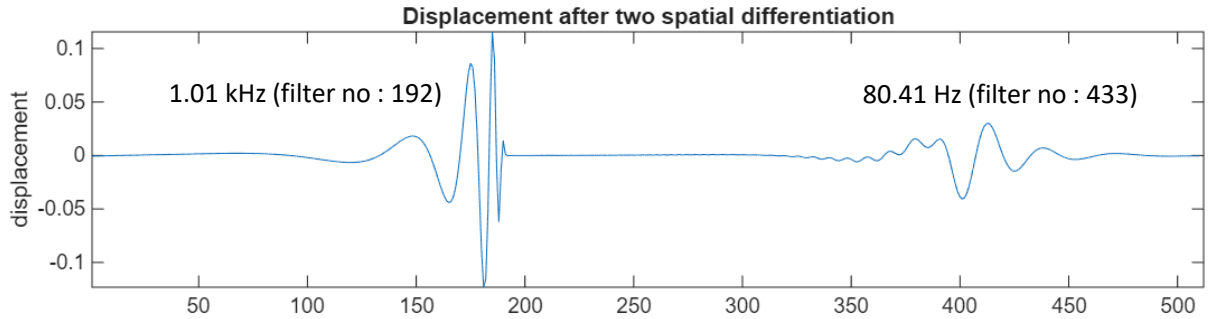
$\tilde{e}[n]$ is the spatially differentiated displacement after half-wave rectification.

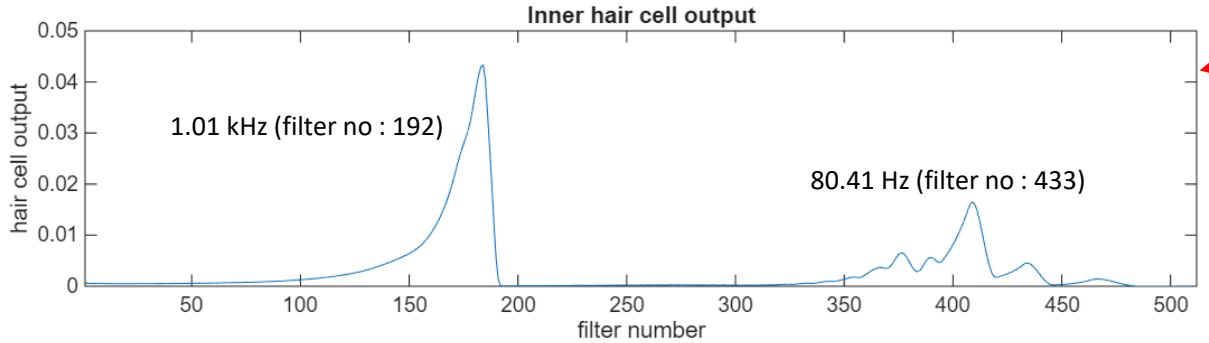
Cut-off frequency (f_c) of the hair cell model is based on the rate at which the switch is closed. (every 16 ms – 62.5 Hz). Therefore $f_c \leq 31.25\ Hz$. For this model, choose f_c=30Hz; Sampling frequency (f_s)=16,000Hz;

Are you on the right track?

- Apply the sum of two sinusoids, at 80.41 Hz and 1.01 kHz, to the input x[n].
- The figure below shows the basilar membrane after two stages of spatial differentiation $\{e_1[n]\ to\ e_N[n]\}$ plotted against the filter number at a particular time instant . The lower plot shows corresponding inner hair cell outputs .
- If your inner hair cell model implementation is correct, you should obtain results similar to those shown below.
- The inner hair cell outputs show two prominent peaks corresponding to the two input frequencies.







Higher-Level Task: Final Validation

- ✓ Generate an input signal containing 1000 to 2000 samples comprising four equal-amplitude sinusoids. Choose distinct frequencies within the filter bank's operating range.
- ✓ After allowing the initial transients to decay, plot the spectrum analyser output (the inner hair cell output in dB) against filter number at a selected time instant. Identify the peaks and their corresponding centre frequencies.
- ✓ Compute and plot the FFT magnitude spectrum of the input signal. Compare the frequencies and relative peak heights with those obtained from the spectrum analyser. Explain any differences, considering filter bandwidth, channel gain and FFT spectral leakage.
- ✓ The input sampling frequency is 16 kHz, while the analyser output is read every 16 ms, corresponding to 62.5 Hz. Explain why this lower output sampling rate is appropriate for tracking slowly varying spectral amplitudes.
- ✓ If the output is read every 8 ms instead (125 Hz), discuss the implications for the hair cell low-pass filter cutoff frequency, smoothing and aliasing. Does the cutoff frequency need to change? Explain your reasoning.
- ✓ Investigate frequency resolution by moving two input frequencies closer together. Can the analyser still distinguish them as separate peaks?

ELEC3104: Mini-Project – Cochlear Signal Processing

Higher –Level (Task 2): Pitch Period Estimation of Voiced Speech Using the Spectrum Analyser

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

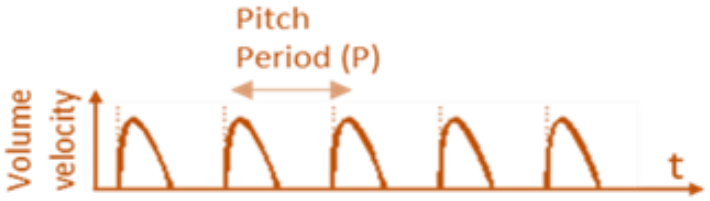
Background to Speech Production

Speech Production

The speech sounds are divided into two broad classes: (a) Voiced Sound, (b) unvoiced sound

(a) Voiced Sound

- VOICED sound (e.g. /a/, /e/, /i/ …) is produced when the vocal cords vibrate during the production of a sound.
- Voiced speech occurs when the air flows through the vocal cords into the vocal tract in discrete "puffs" (as shown on the figure) rather than as a continuous flow.

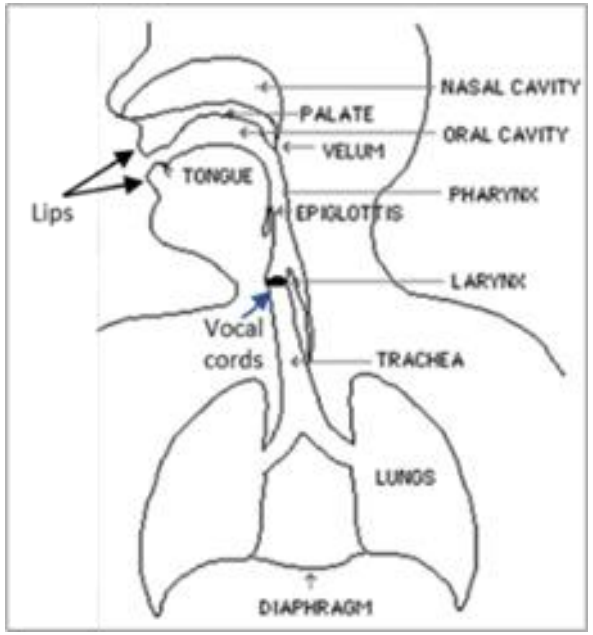


- The vocal cords vibrate at a particular frequency, which is called the fundamental frequency (defined as pitch and the pitch period = P) of the voiced sound. The fundamental frequencies are between:
  - ✓ 50 - 200 Hz for male speakers
  - ✓ 150 - 300 Hz for female speakers
  - ✓ 200 - 400 Hz for child speakers

(b) Unvoiced Sound

- UNVOICED speech (e.g. /s/, /f/…) is produced when vocal cords are not vibrating and they are held open and the air flows continuously through them. Unvoiced speech is characterised by high frequency components.

Speech Production Mechanism



Background to Speech Production

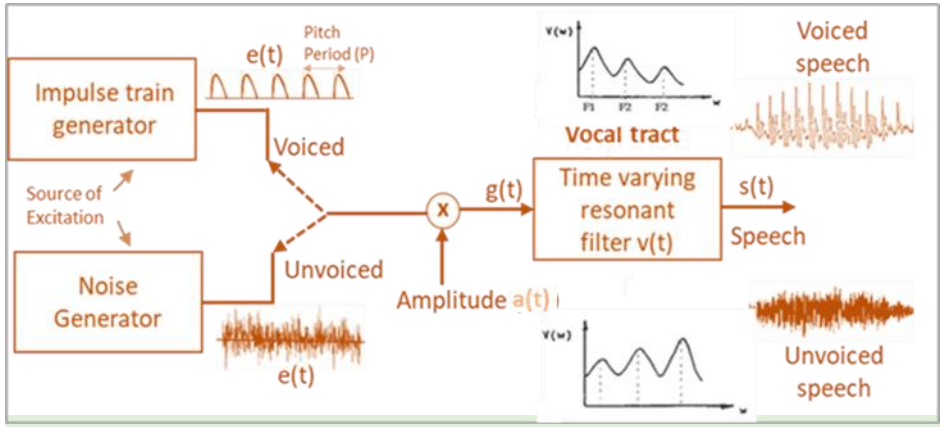
Vocal Tract

- Vocal tract is a non-uniform acoustic tube that is terminated at one end by the vocal cords and at the other end by the lips;
- The vocal tract changes shape rather slowly in continuous speech, and it is reasonable to assume that the vocal tract has a fixed characteristic over a time interval of the order of 10 – 20 ms;
- Thus, once every 10 -20 ms, the vocal tract configuration is varied, producing new vocal tract parameters (resonant frequencies). The spectrum of vocal tract response consists of a number of resonant frequencies of the vocal tract.

Speech Production Model

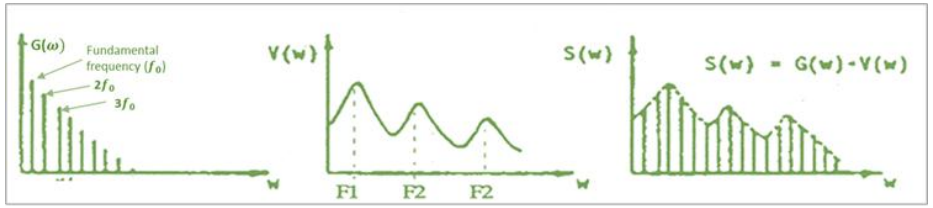
- The spectrum of vocal tract response consists of a number of resonant frequencies (F1, F2, F3, etc.) of the vocal tract and these frequencies are called Formants.
- Three to four formants present below 4kHz of speech. The speech signal is a slow time varying signal in the sense that when examined over sufficiently short period of time, its characteristics are stationary.
- Thus, once every 10 – 20 ms, on average, the vocal tract configuration is varied producing new vocal tract parameters (resonant frequencies, F1, F2, F3 etc.). A simplified speech production model is shown on the right:

Simplified speech production model

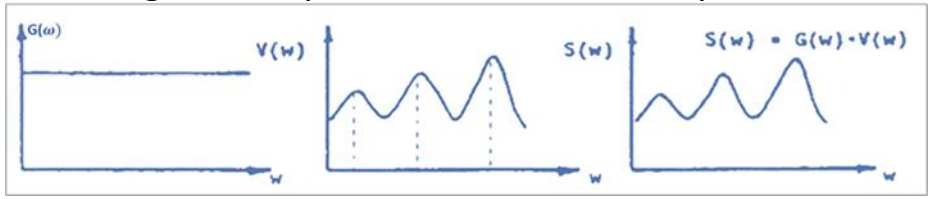


$$g(t) = a(t).e(t)\,; \quad s(t) = g(t) * v(t)$$
$$S(\omega) = G(\omega).V(\omega)$$

Magnitude Spectrum of Voiced Speech



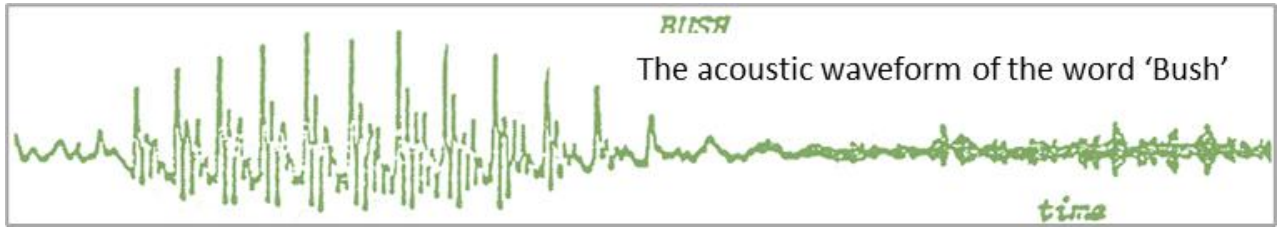
Magnitude Spectrum of Unvoiced Speech



Background to Speech Production

- The nature of the excitation function e[n] of the model will be different, depending on the nature of the speech sounds being produced.
- For voiced speech, the excitation e[n] will be a train of unit impulses spaced at intervals of the pitch period ($e[n]=\delta[n-Pk]$ k=0,1,2…).
- For unvoiced speech, the excitation will be a random noise-like signal ($e[n]=random[n]$). For either voiced or unvoiced speech, the underlying spectrum of the vocal tract will exhibit distinct frequency peaks.
- For VOICED speech, the magnitude of the lower formant frequencies is successively larger than the magnitude of the higher formant frequencies.
- For UNVOICED speech, the magnitude of the higher formant frequencies is successively larger than the magnitude of the lower formant frequencies.

For example, consider the spoken word 'bush' and the sound /a/. 'Bush' has parts that are voiced and parts that are not (unvoiced). The sound /a/ is a voiced and you can easily see its pitch period in the diagram.





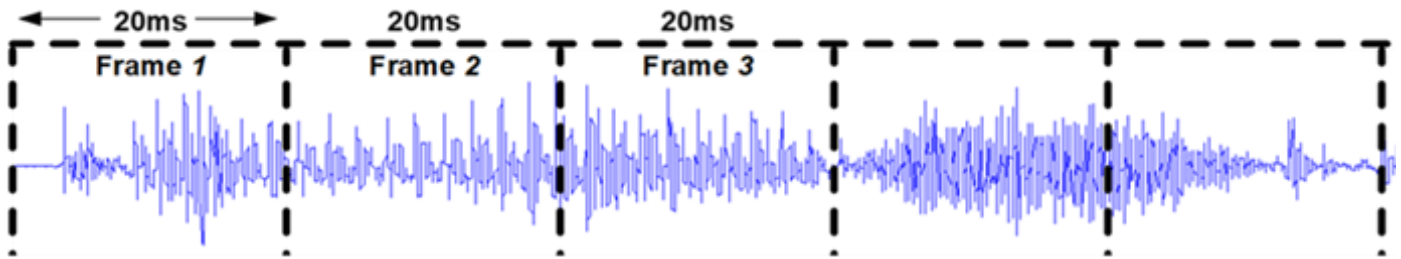
Reference:

https://thebox.unsw.edu.au/video/speech-and-audio-processing-1-introduction-to-speech-processing-professor-e-ambikairajah

https://thebox.unsw.edu.au/video/speech-and-audio-processing-2-speech-analysis-professor-e-ambikairajah

Pitch Period Estimation in Time domain

- For Task 2, record your own voice at a sampling frequency of 16 kHz (the voice signal bandwidth is 0–8 kHz). The objective is to determine the pitch period (or the inverse of the fundamental frequency) of your voice.
- The sentence to be spoken is: "**We were away.**" Say the entire sentence without any pauses and save the recorded sound file.
- Before estimating the pitch period, listen to your own voice using the MATLAB *soundsc* command, and display the waveform on the computer screen. Show this waveform to your lab demonstrator.
- Since the vocal tract configuration changes every **10 –20 ms**, divide the voiced speech into non-overlapping 20 ms (320 samples) frames (see figure below).
- To find the pitch period (or fundamental frequency), manually count the number of samples between similar peaks **within one frame**, and repeat this process for **another frame**. Check if the fundamental frequency is the same for both frames . Make sure you know how to calculate the fundamental frequency (pitch) from the number of samples. Discuss your results with your lab demonstrator.



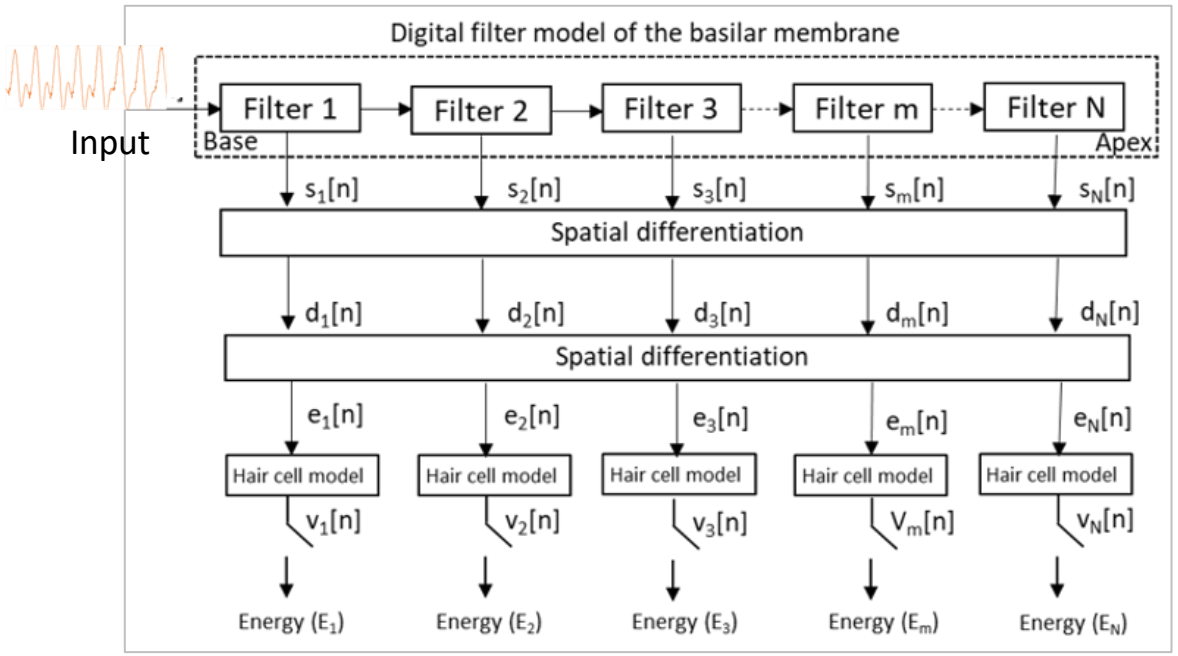
Voiced speech

Pitch Period Estimation Using a Spectrum Analyser

1. For this task, use the spectrum analyser that you implemented and tested in Task 1.
2. Use your saved speech signal, $s(n)$, **'We were away'**, as the input to the spectrum analyser (diagram on the right).
3. Observe the energy output of your spectrum analyser for a single frame 20ms .
4. It is assumed that the fundamental frequency of the input speech signal (Male or Female) lies in the range of 90 Hz (Filter 422) to 401 Hz (Filter 280),
5. The inner hair-cell output of each filter and the outputs of the filters corresponding to its Hth harmonics are multiplied together to obtain the harmonic product spectrum (HPS):
6. $$HPS_i = \prod_{k=1}^{H} v_{ik}$$

   - i: filter number, from 280 to 422.
   - H: number of harmonics; use $H = 4$ as specified in Table 1.
   - v_{ik}: hair-cell output corresponding to the kth harmonic for filter i.
7. Calculate HPS_i for all 143 filters in the fundamental-frequency range of 90–401 Hz.
8. Use Table 1 (see next slide) to identify the filters corresponding to the fourth harmonics of the 143 centre-frequency filters, then calculate HPS_i for those 143 filters.

Spectrum Analyser



Ouput Energy E_m is read (switch closed) once every 20 ms or so.

9. Plot HPS_i against filter number for filters 280–422.
10. Apply a peak-picking algorithm to locate the highest peak. The corresponding filter number indicates the fundamental frequency of the voiced speech frame.

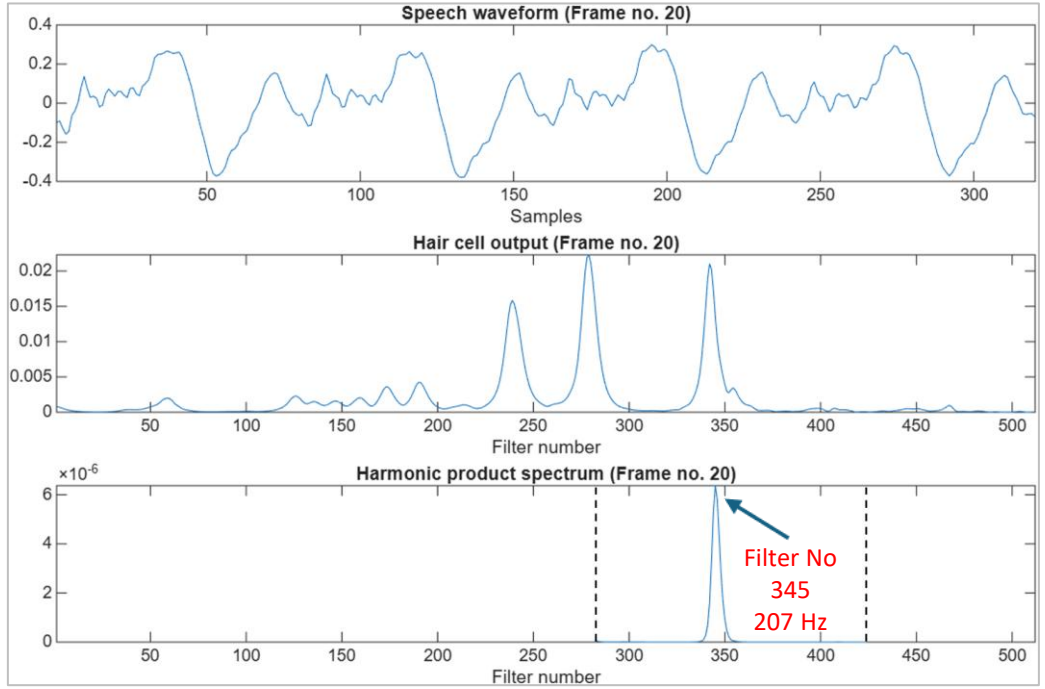
Table 1: Fundamental frequency (f_0) and 2nd to 4th harmonics of the filters 280 to 422

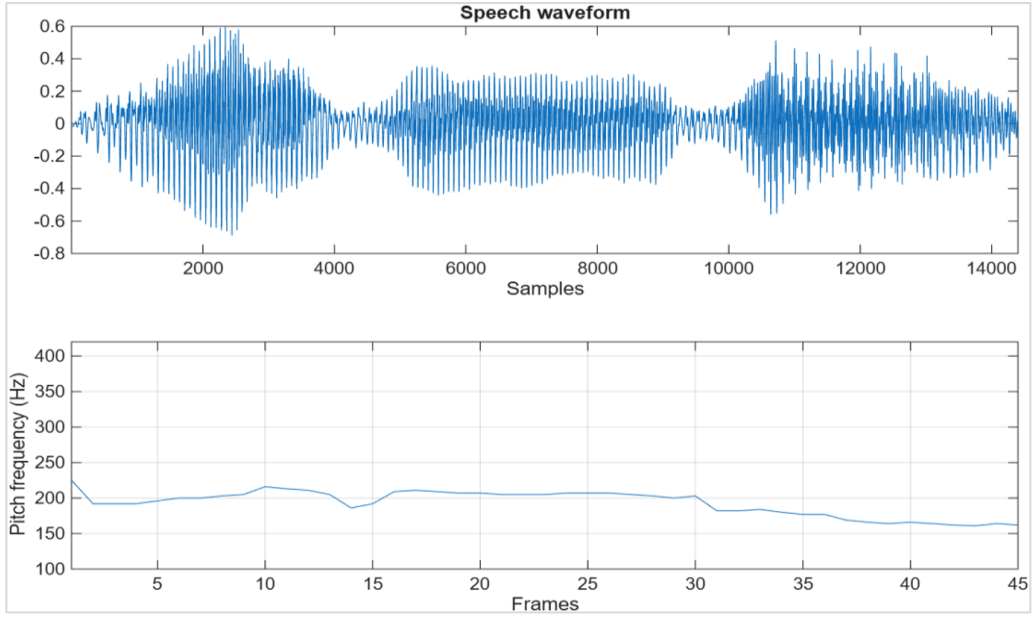
Filter number	Fundamental Frequency (Hz) f_0	2nd Harmonic $2f_0$	3rd Harmonic $3f_0$	4th Harmonic $4f_0$
422	90 Hz	356	318	290
421	91	355	317	289
420	92	354	316	288
419	93	353	315	287
418	94	352	314	286
-	-	-	-	-
-	-	-	-	-
281	397	215	177	149
280	401 Hz	214	175	148

The full table is available in the **Mini-Project** folder on Moodle.

Are you on the right track?

- ✓ The figure on the right shows a single 20 ms frame (320 samples) of voiced speech, its corresponding hair-cell output, and the harmonic product spectrum (HPS).
- ✓ From the hair-cell output alone, it is difficult to identify the fundamental frequency accurately.
- ✓ Applying the harmonic product spectrum (HPS) enhances the peak corresponding to the fundamental frequency while suppressing peaks associated with other frequencies. The location of the highest peak (filter number) is used to estimate the fundamental frequency of the voiced speech frame.





- ✓ The figure on the left shows the speech waveform and the estimated fundamental frequency for a female speaker. The fundamental frequency is estimated using the harmonic product spectrum (HPS) for each 20 ms frame.
- ✓ The spoken utterance is utterance, “We were away”, which consists entirely of voiced frames, making pitch estimation possible throughout the utterance.
- ✓ Download the files Female16K.wav from the **Mini-Project** folder on Moodle. Implement your HPS algorithm and compare your results with those shown in the figure on the left.

ELEC3104: Mini-Project – Cochlear Signal Processing

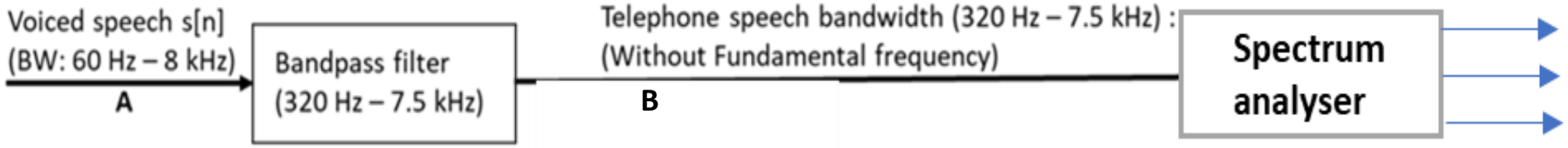
Higher –Level (Task 3): Pitch Estimation When the Fundamental Frequency is Absent

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

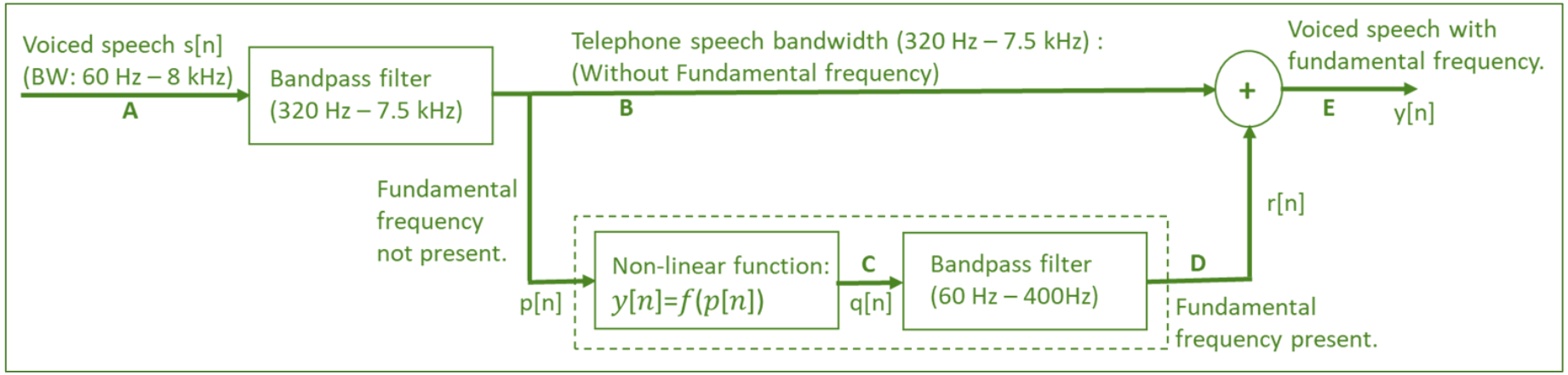
Pitch Estimation When the Fundamental Frequency is Absent

- Typically, telephone speech doesn't include frequencies up to 320 Hz, meaning the speaker's fundamental frequency is absent.
- Your task is to design a band-pass filter with a lower cut-off frequency of 320 Hz and an upper cut-off frequency of 7.5 kHz.
- When you apply this filter to your recorded speech file, the resulting output won't contain any fundamental frequencies. Listen to the original speech (at A - see figure below) and the filtered speech (at B - see figure below). Do you notice any differences in the quality? Discuss with your lab demonstrator.
- Now, give the filtered speech as input to your spectrum analyser. Select a frame of your choice and observe the energy corresponding to the filter that was previously identified as the pitch frequency for that frame. When the fundamental component is absent, the energy in this band will be very low. Repeat the process for another frame and discuss your observations with the lab demonstrator, comparing the results with those from Task 2 and Task 3.



Auditory Pitch Perception: Regeneration of the Missing Fundamental

- When the fundamental frequency is missing (as in band-limited or telephone speech), people can still perceive the pitch - a phenomenon known as the missing fundamental. Pitch perception is influenced by the harmonic relationships of the fundamental frequency present in the speech signal.
- One theory of auditory pitch perception suggests that nonlinearities in the auditory system (either in the middle ear or the inner ear) can recreate the missing fundamental frequency from the existing harmonics.
- In this task, you will explore this theory using your filtered speech file. If you pass the filtered speech through a nonlinear circuit followed by a band-pass filter (see diagram below), the fundamental frequency can be regenerated from its harmonic components.
- Write a MATLAB program to estimate the pitch period at point E (Task 4), as shown in the diagram below, using your spectrum analyser. Examine the spectrogram (or the magnitude spectrum for each frame) at different positions (A, B, C, D, and E) using the FFT magnitude spectrum, and draw conclusions based on your observations
- Finally, discuss your results (regeneration Succeeds or fails) with the lab demonstrator.



Higher-Level Task: Final Validation

- ✓ Choose a sampling frequency of 16 kHz. Select a fundamental frequency $f_0 < 200\ Hz$ that is also the centre frequency of one of the analysis filters below 200 Hz. Write MATLAB code to:

    1. Generate a sinusoid at $f_0$with amplitude 1 and length 2000 samples.
    2. Generate the next eight harmonics ($2f_0, 3f_0, \ldots, 9f_0$,(each 2000 samples long, with decreasing amplitudes.
    3. Add all sinusoids together to form one synthetic signal (2000 samples total).

- ✓ Use this synthetic signal as the input to Task 1, Task 2 and Task 3. The combined signal should go directly into the spectrum analyser.

- ✓ Verify that your pitch estimation based on the missing fundamental still identifies f_0 , (Task 3) even though f_0 may not be present with high energy.

- ✓ Discuss your results with your lab demonstrator.

Reflection: Higher-Level Tasks 1, 2 and 3

Support your reflection with specific results and plots from your project.

- **Spectrum analysis:** How did your cochlear spectrum analyser compare with the FFT magnitude spectrum? Explain any differences in peak frequencies, relative amplitudes and frequency resolution.
- **Inner hair cell model:** What effects did half-wave rectification and the 30 Hz low-pass filter have on the analyser output? Why was a 30 Hz cutoff frequency chosen, and how does it affect the smoothing and time response of the output? What changes when the output reading interval is reduced from 16 ms to 8 ms? Explain your observations.
- **Pitch period estimation:** How closely did your HPS-based pitch estimates agree with your manual measurements from the speech waveform? Explain any differences.
- **Harmonic Product Spectrum (HPS):** How did combining the outputs corresponding to the fundamental frequency and its harmonics help identify the pitch? What effect did using four harmonics have on the estimated fundamental frequency?
- **Missing fundamental:** What happened to the perceived pitch and the analyser output after band-pass filtering? Could the HPS still estimate the fundamental frequency from the remaining harmonics? Explain your observations.
- Learning and validation: Describe one unexpected result or implementation error encountered during the project. Explain how you investigated the problem, what you changed (if anything), and what you learned. If you used GenAI, give one example of a suggestion and explain how you independently verified it.
- **Improvements:** What would you change to improve your system, and how would you test the improvement?